Zhonghao Zhang

zhonghao.zhang@vanderbilt.edu | (615) 484-7413 |

EDUCATION

Vanderbilt University Nashville, TN
Bachelor of Engineering in Electrical and Computer Engineering & Mathematics *May 2023 – May 2027*
Minor in Computer Science

- **GPA: 3.85 / 4.0**

- **Relevant Coursework:** Robotics, Control Systems, Applied Machine Learning, Found of Machine learning, Image Processing, Computer Vision, Intermed Software Design, Digital Signal Processing, Operating System, Linear Algebra

PUBLICATIONS

[1] **Zhang, Z.***, Wu, M.*, Yang, H., Acar, A., Kuntz, A., and Wu, J. Y. "Object-Centered Reconstruction for Vision-Based 3D Force Estimation." Submitted to the *IEEE International Conference on Robotics and Automation* (ICRA), 2027.

[2] Wu, M.*, **Zhang, Z. ***, Yang, H., Kuntz, A., and Wu, J. Y. "Transferable Tool-Tissue Contact Detection from Stereo Depth in Robot-Assisted Surgery." Submitted to *SPIE Medical Imaging*, 2027. (*Co-first authors)

[3] **Zhang, Z.**, Zhou, H., Yang, H., Fischer, G. S., and Kazanzides, P. "Dynamics Estimation for the da Vinci Research Kit: A Review of Model-Based and Learning-Based Approaches." *Machines*, vol. 14, no. 8, p. 879, 2026. DOI: 10.3390/machines14080879

[4] **Zhang, Z.**, Lu, M., Lu, Z., Huo, Y., and Yan, X. "Reducing RF-Induced Heating of DBS in 3 T MRI Using Dual-Role Receive Arrays as Wireless Resonators: A Simulation Study." *IEEE Access*, vol. 14, pp. 112236–112247, 2026. DOI: 10.1109/ACCESS.2026.3713428

[5] **Zhang, Z.**, Lu, M., Liang, H., Zu, Z., Gu, Y., Wang, X., Huo, Y., and Yan, X. "Fast Electromagnetic and RF Circuit Co-Simulation for Passive Resonator Field Calculation and Optimization in MRI." *Magnetic Resonance Imaging*, vol. 129, p. 110644, 2026. DOI: 10.1016/j.mri.2026.110644

[6] Yang, H., **Zhang, Z.**, Al-Zogbi, L., Kazanzides, P., and Wu, J. Y. "Friction Compensation for Improved Sensorless Force Estimation in the dVRK-Si." Submitted to *IEEE Transactions on Medical Robotics and Bionics*, 2026.

[7] Zhou, H., Yang, H., Burkhart, B., Deguet, A., **Zhang, Z.**, Fichera, L., Fischer, G. S., Wu, J. Y., and Kazanzides, P. "Towards Dynamic Model Identification and Gravity Compensation for the dVRK-Si Patient Side Manipulator." *IEEE Transactions on Medical Robotics and Bionics*, accepted 2026.

[8] **Zhang, Z.**, Lu, M., Wang, X., Huo, Y., and Yan, X. "Electromagnetic Field Optimization for Safe MRI of Patients with Deep Brain Stimulation Using Wireless Coils." Abstract accepted at the ISMRM & ISMRT Annual Meeting and Exhibition, Cape Town, South Africa, 2026.

[9] **Zhang, Z.,** Wu, M., Yang, H., Acar, A., Kuntz, A., and Wu, J. Y. "Stereo Vision-Based Force Estimation and 3D Reconstruction Toward Autonomous Colorectal Anastomosis: A Preliminary Study." Submitted to the 2nd International Workshop on the Evolving Landscape of Surgical Robotics (ELSR), held in conjunction with the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

PROFESSIONAL EXPERIENCE

Machine Automation, Perception and Learning (MAPLE) Lab, Vanderbilt University Nashville, TN
Research Assistant, advised by Prof. Jieying Wu *Jan. 2026 – Present*

- Developed a vision-based force estimation pipeline for the da Vinci Surgical System Si, integrating stereo depth estimation with SAM2 segmentation to reconstruct a 3D tissue point cloud, Chamfer-loss registration with local linear regularization, and a Set Transformer–LSTM model for force prediction
- Built a vision-based contact detection pipeline using SAM2 segmentation and FoundationStereo depth estimation, with a two-state hidden Markov model incorporating temporal features for robust contact classification
- Implemented sensorless force estimation on the dVRK-Si by identifying robot kinematics and dynamics parameters for gravity compensation and computed-torque feedforward control

- Developed a C++ stereo video acquisition pipeline with ROS integration for synchronized data collection

RF Laboratory, Vanderbilt University Nashville, TN
Research Assistant, advised by Prof. Xinqiang Yan *Dec. 2024 – Feb. 2026*

- Designed 2-, 6-, and 32-element wireless RF coil models in ANSYS HFSS for 3T MRI and developed a fast electromagnetic co-simulation algorithm reducing computation time from hours to milliseconds
- Built a physics-informed deep learning framework in PyTorch integrating EM co-simulation, ResNet-18, and genetic algorithms to optimize RF coil parameters for reducing heating in DBS implants and improving field homogeneity

Robotics and Autonomous Systems Laboratory, Vanderbilt University Nashville, TN
Research Assistant, advised by Prof. Nilanjan Sarkar *Aug. 2024 – Dec. 2024*

- Developed an embodied AI assistant for dementia care, using LLMs to generate context-aware non-verbal gestures and driving a 3D avatar in Unity for real-time human-AI interaction, with caregiver feedback integrated to improve emotional appropriateness

Intelligent Sensor and Detection Technology Laboratory, Shandong University Shandong, China
Research Assistant, advised by Prof. Mingshun Jiang *May 2024 – Aug. 2024*

- Built a real-time sensor data acquisition and visualization system using Java socket communication over Ethernet, with digital signal processing pipelines for noise reduction and a PyQt GUI for live monitoring and analysis

Rock robot Shanghai, China
Engineer Assistant *Apr. 2023 – Jul. 2023*

- Designed and optimized infrared camera mount structures in SolidWork, and implemented a Python-based computer vision heat source detection pipeline using Opencv for thermal imaging applications

TECHNICAL SKILLS

- **Programing Language:** Python (advanced), Matlab(advanced), Java(advanced), C++(advanced)
- **Software and Tools:** ANSYS HFSS, SolidWorks, LTspice